Final Project Submission

DeepCompile: AI-Driven Compiler Optimization

Roll Number: 24CSB0A06 · Date: April 19, 2026

Comparative Analysis Report

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Compared Variants

- Original baseline 4-stage pipeline
- Revised orchestrated deterministic workflow

Aggregate Comparison

- Baseline `success` count: **10** / 12
- Orchestrated `PASS` workflow count: **10** / 12
- Cases improved by orchestration/refinement: **0**
- Average latency delta (`baseline - orchestrated`): **-4.5528 s**

Case-by-Case Comparison

File	Baseline Status	Orchestrated Workflow	Delta (s)	Stop Reason
`TC01_uninit_arithmetic.cpp`	success	PASS	-10.8223	initial_pass
`TC07_div_by_zero_var.cpp`	success	PASS	-14.1112	initial_pass
`TC11_buffer_overflow_loop.cpp`	success	PASS	-2.8706	initial_pass
`TC12_strcpy_overflow.cpp`	success	PASS	-3.4444	initial_pass
`TC14_memory_leak.cpp`	success	PASS	-2.9276	initial_pass
`TC16_use_after_free.cpp`	rollback	ESCALATE	-3.0772	no_material_improvement
`TC20_leak_in_loop.cpp`	success	PASS	-2.6597	initial_pass
`TC27_int_overflow.cpp`	success	PASS	-2.787	initial_pass
`TC31_off_by_one.cpp`	rollback	ESCALATE	-3.0539	no_material_improvement
`TC35_unreachable_code.cpp`	success	PASS	-3.0447	initial_pass

| `TC39_empty_if_body.cpp` | success | PASS | -3.0039 | initial_pass |

| `TC41_leak_early_return.cpp` | success | PASS | -2.8312 | initial_pass |

Interpretation

- The orchestrated workflow adds explicit bounded refinement and escalation metadata.
- Comparisons here stay within the revised project scope: source-to-source workflow only.
- No compiler baseline or LLM-only baseline is used in this formal report set.